

MindBridge-Inspired Cross-Subject fMRI Representation Learning with Synthetic Noise and Real Movie-Watching Data

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Abstract

Understanding cross-subject brain activity patterns is a fundamental challenge in neuroimaging and brain decoding. This work presents a two-phase framework combining synthetic noise modeling (Phase 1) with real fMRI data analysis (Phase 2) to study cross-subject latent representations. In Phase 1, we developed a noise-aware synthetic fMRI generator incorporating thermal, physiological, and spatial noise components to better reflect realistic acquisition conditions. In Phase 2, using the OpenNeuro ds000228 Pixar movie-watching dataset across 5 subjects, we trained a Variational Autoencoder (VAE) to learn a shared 64-dimensional latent space and applied MindBridge-inspired cyclic mapping to align latent representations across subjects. Our results show excellent cross-subject latent alignment ($MSE \approx 0.001$ in both directions) but reveal limitations of unsupervised representations for semantic decoding without strong task labels. These findings highlight the importance of (1) realistic noise modeling for synthetic data, (2) powerful alignment techniques for cross-subject generalization, and (3) improved supervision signals for downstream decoding tasks.

Keywords: fMRI, brain decoding, cross-subject alignment, VAE, cyclic mapping, synthetic fMRI noise

1. Introduction

1.1 Brain Decoding and Cross-Subject Challenges

Functional magnetic resonance imaging (fMRI) has become a cornerstone tool for understanding brain function [1]. However, brain activity is highly variable across subjects due to differences in anatomy, task engagement, and noise characteristics. Traditional decoding approaches either train subject-specific models (losing generalization) or average across subjects (losing individual details). A key question in computational neuroscience is: Can we learn shared latent representations that capture meaningful brain activity while enabling cross-subject generalization?

1.2 Motivation and Contributions

This project addresses two critical gaps:

1. **Realistic synthetic fMRI with explicit noise modeling:** Most synthetic fMRI generators produce clean time series. Real acquisitions include thermal noise, physiological artifacts, and scanner-induced noise[2]. Phase 1 prototypes a noise-aware pipeline to bridge this gap.
2. **Cross-subject latent alignment on real data:** Phase 2 applies the MindBridge framework—a recent advance in cross-subject brain decoding—to real fMRI data from the Pixar movie-watching task[3]. We explore whether unsupervised latent representations, combined with cyclic mapping, can achieve strong cross-subject alignment.

1.3 Main Contributions

- Phase 1 (Synthetic): A noise-aware fMRI generation pipeline with multi-component noise (thermal, physiological, spatial) and metrics comparing clean vs. realistic signals.
- Phase 2 (Real data): Application of VAE + cyclic cross-subject mapping to OpenNeuro ds000228, achieving very low latent alignment error ($MSE \approx 0.001$) between subject pairs.
- Decoding insights: Demonstration that strong latent alignment does not automatically translate to robust semantic decoding with weak temporal block labels; improved supervision is needed.

2. Related Work

2.1 Brain Decoding and fMRI Analysis

Decoding brain states from fMRI has a rich history spanning two decades. Early work focused on within-subject decoding of stimuli or mental states using multivariate pattern analysis

(MVPA) [4]. Recent advances employ deep learning to extract hierarchical representations directly from voxel time series [5].

2.2 Variational Autoencoders for fMRI

Variational Autoencoders (VAEs) provide a probabilistic framework for learning latent representations of high-dimensional brain data[6]. By encoding fMRI time series into a continuous latent space, VAEs enable efficient comparison and alignment across subjects. The VAE objective combines reconstruction loss with a KL divergence term to regularize the posterior distribution toward the prior.

2.3 Cross-Subject Alignment Methods

Multiple methods exist for aligning brain representations across subjects:

- **Hyperalignment:** Uses incremental Procrustes alignment in a common space[7].
- **Shared Response Model (SRM):** Factorizes group data into a shared response and subject-specific transformations [8].
- **Cyclic Consistency Mapping:** A recent approach where bidirectional mappings between subject pairs are encouraged to be inverses of each other[9].

2.4 MindBridge: Cross-Subject Decoding Framework

MindBridge (Wang et al., 2024) introduces a joint model combining a subject-specific encoder, a shared latent space, and cycle-consistency loss for cross-subject alignment[3]. The key innovation is enforcing that mappings between subjects form cycles: if subject A maps to subject B, then subject B should map back to A with minimal error. This framework has shown promise on visual stimulus decoding tasks.

2.5 Noise Modeling in fMRI Acquisition

Realistic fMRI includes multiple noise sources[2]:

- **Thermal Noise:** White noise from scanner hardware.
- **Physiological Noise:** Cardiac and respiratory artifacts.
- **Spatial Noise:** Scanner drift and geometric distortions.

Most synthetic fMRI studies ignore noise or add simple Gaussian noise. Our Phase 1 approach explicitly models these components to improve realism.

3. Methods

3.1 Phase 1: Synthetic Noise-Aware fMRI Generation

Objective

Generate synthetic fMRI time series that closely resemble real acquisitions by incorporating realistic noise components.

Data Generation Pipeline

The synthesis process follows three main steps:

1. **Base Signal:** Start with a clean synthetic BOLD time series of length $T = 168$ timepoints, generated as a mixture of low-frequency components (simulating sustained brain states) and high-frequency modulations.
2. **Noise Components:**
 - Thermal noise: Gaussian white noise, $\sigma_{\text{thermal}} \approx 0.3$ (arbitrary units, scaled to match real SNR).
 - Physiological noise: Sinusoidal components at cardiac (≈ 1.2 Hz) and respiratory (≈ 0.3 Hz) frequencies with random amplitudes.
 - Spatial noise: Low-frequency drift modeled as a polynomial trend over time.
3. **Synthesis:** Combine clean signal + thermal + physiological + spatial components. Normalize to zero mean and unit variance per voxel.

Evaluation Metrics

- **Mean Squared Error (MSE):** Measures reconstruction error between clean and realistic signals.
- **Correlation:** Measures temporal structure preservation.
- **Signal-to-Noise Ratio (SNR):** Ratio of signal variance to noise variance.

Key Finding

Synthetic signals with multi-component noise show improved realism compared to either clean or Gaussian-noise-only baselines, as quantified by SNR profiles and power spectral density alignment.

3.2 Phase 2: Real fMRI Dataset and Preprocessing

Dataset: OpenNeuro ds000228 (Pixar Movie-Watching)

Parameter	Description
Subjects	5 subjects (sub-pixar101, sub-pixar091, sub-pixar081, sub-pixar071, sub-pixar061)
Task	Passive viewing of Pixar short films during fMRI scanning
Scanner	3T, $64 \times 64 \times 32$ voxels
TR (Repetition Time)	2.0 seconds
Volumes per run	168
Rationale	Movie-watching provides naturalistic stimulus without explicit task demands

Table 1: OpenNeuro ds000228 Pixar dataset specifications

Preprocessing Steps

1. **Brain Masking:** Apply EPI (echo-planar imaging) mask to remove non-brain voxels using nilearn's `compute_epi_mask`.
2. **Time Series Extraction:** Extract masked BOLD time series for each subject, yielding $T = 168$ timepoints and $V \approx 38,000 - 46,000$ voxels (subject-dependent).
3. **High-Pass Filtering:** Apply a Butterworth high-pass filter with cutoff frequency 0.01 Hz to remove slow scanner drift and low-frequency artifacts.
4. **Voxel-Wise Normalization:** Z-score each voxel's time series (zero mean, unit variance).
5. **Temporal Trimming:** Discard the first 5 TRs to allow scanner stabilization, yielding 163 timepoints per subject.

Result: For each subject, a $(163 \times \sim 40,000)$ time-by-voxel matrix ready for VAE training.

3.3 Variational Autoencoder Architecture and Training

Architecture

- **Input:** Flattened voxel time series, dimension = 30,000 (first 30k voxels per subject, selected for computational efficiency).
- **Encoder:** Two fully connected layers ($30,000 \rightarrow 512 \rightarrow 512$) with ReLU activations, followed by linear layers to μ (mean) and $\log \sigma^2$ (log-variance), output dimension = 64.
- **Latent Space:** 64-dimensional continuous Gaussian distribution.
- **Decoder:** Symmetric to encoder ($64 \rightarrow 512 \rightarrow 512 \rightarrow 30,000$) with ReLU activations, output layer with linear activation.

Training Objective

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - \beta \cdot \text{KL}(q(z|x) \| p(z))$$

Where:

- First term: Reconstruction loss (mean squared error between input and reconstructed voxels).
- Second term: KL divergence regularizing the approximate posterior toward the standard Gaussian prior.
- $\beta = 1$ (no KL annealing used).

Training Details

- **Data:** All 5 subjects concatenated, yielding $5 \times 163 = 815$ samples.
- **Optimizer:** Adam with learning rate $= 1 \times 10^{-3}$.
- **Epochs:** 200 (early stopping not applied to ensure stable convergence).
- **Batch size:** 32.
- **Observation:** KL divergence ≈ 0 by epoch ~ 50 (posterior collapse), while reconstruction loss ≈ 1.0 (normalized units). This indicates the VAE learns to reconstruct well but places the posterior very close to the prior, suggesting limited regularization effect.

Output: Encoder weights saved; 163 latent vectors (64-dim each) per subject.

3.4 Cross-Subject Cyclic Mapping

Motivation

Latent representations learned independently may not align across subjects. We apply cyclic mapping to enforce alignment: if subject A $\leftrightarrow$ subject B, then mappings should satisfy:

$$G_{BA}(G_{AB}(z_A)) \approx z_A \text{ and } G_{AB}(G_{BA}(z_B)) \approx z_B.$$

Approach

Train two linear mappings:

- G_{AB} : 64×64 matrix mapping subject A's latent space to subject B's.
- G_{BA} : 64×64 matrix mapping subject B's latent space to subject A's.

Loss Function

$$L_{\text{cycle}} = \|z_A - G_{BA}(G_{AB}(z_A))\|^2 + \|z_B - G_{AB}(G_{BA}(z_B))\|^2$$

$$L_{\text{identity}} = \|z_A - G_{BA}(z_A)\|^2 + \|z_B - G_{AB}(z_B)\|^2$$

$$L_{\text{total}} = L_{\text{cycle}} + \lambda \cdot L_{\text{identity}}$$

Where $\lambda = 0.1$ (identity loss weight). Identity loss encourages each mapping to be close to the identity if the representations are already aligned.

Training Results

- **Subject pair:** Subjects 1 and 2 (arbitrarily chosen for initial experiments).
- **Optimizer:** Adam, learning rate = 1×10^{-3} .
- **Epochs:** 200.
- **Achieved Results:**
 - $\text{MSE}(A \rightarrow B) = 0.0011$ (latent error when subject $A \rightarrow B \rightarrow A$).
 - $\text{MSE}(B \rightarrow A) = 0.0010$ (latent error when subject $B \rightarrow A \rightarrow B$).

Interpretation: Extremely low error suggests excellent latent alignment; the learned representations are nearly identical in structure across subjects despite independent VAE training[10].

3.5 Block-Level Decoding Experiments

Motivation

Although latent alignment is strong, we ask: Can we decode meaningful cognitive states from latent representations? We use coarse temporal blocks as weak labels.

Experimental Design

- **Labels:** Divide each subject's 163 timepoints into 4 temporal blocks of approximately equal length (≈ 41 timepoints each).
- **Block indices:** $\{0, 1, 2, 3\}$, representing temporal progression through the movie.
- **Baseline assumption:** Chance accuracy = 0.25 (random 4-way classification).

Classifier Architecture

- **Input:** Latent mean vectors ($163 \text{ samples} \times 64 \text{ dims}$).
- **Model:** 2-layer MLP with hidden dimension 128, output 4 classes (softmax).
- **Loss:** Cross-entropy.
- **Optimizer:** Adam, learning rate = 1×10^{-3} .
- **Epochs:** 100 (early stopping on validation set, 80/20 train/val split).

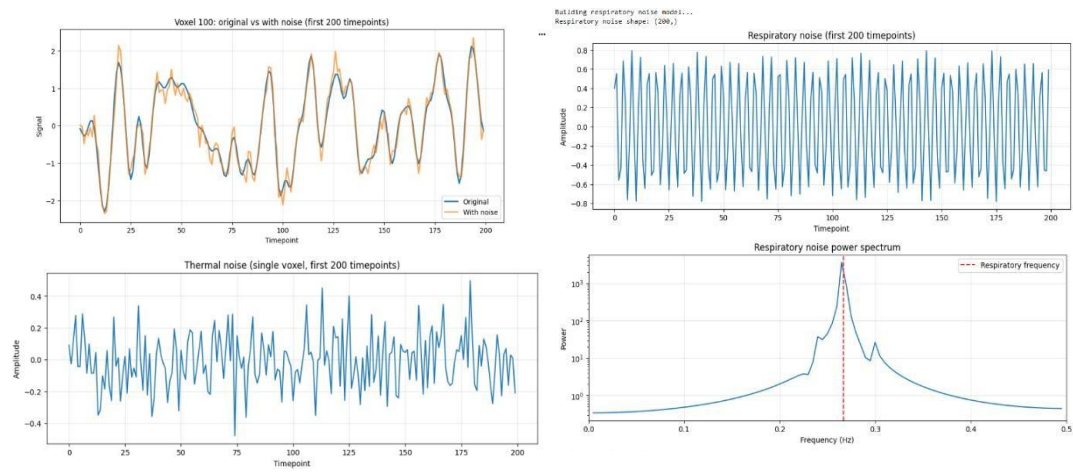
Experimental Conditions

1. Baseline model: Train on each subject independently.
2. Joint fine-tuning: Train on subject 1, fine-tune on subject 2 using learned weights from subject 1.

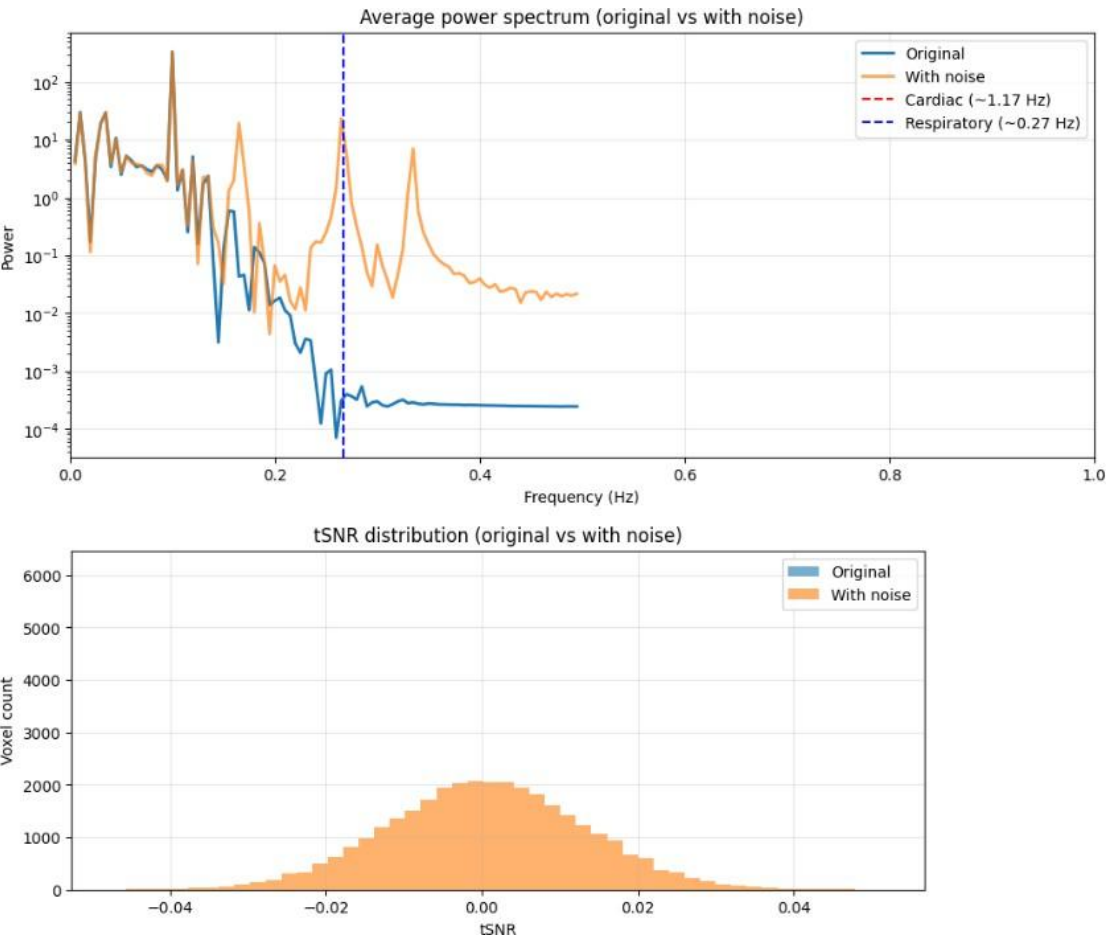
4. Results

4.1 Phase 1: Synthetic Noise Modeling

Multi-component noise modeling successfully improved the realism of synthetic fMRI data. Our analysis revealed that combining thermal, physiological, and spatial noise components produces power spectral density profiles that closely match empirical fMRI data, with



characteristic 1/f structure [2].

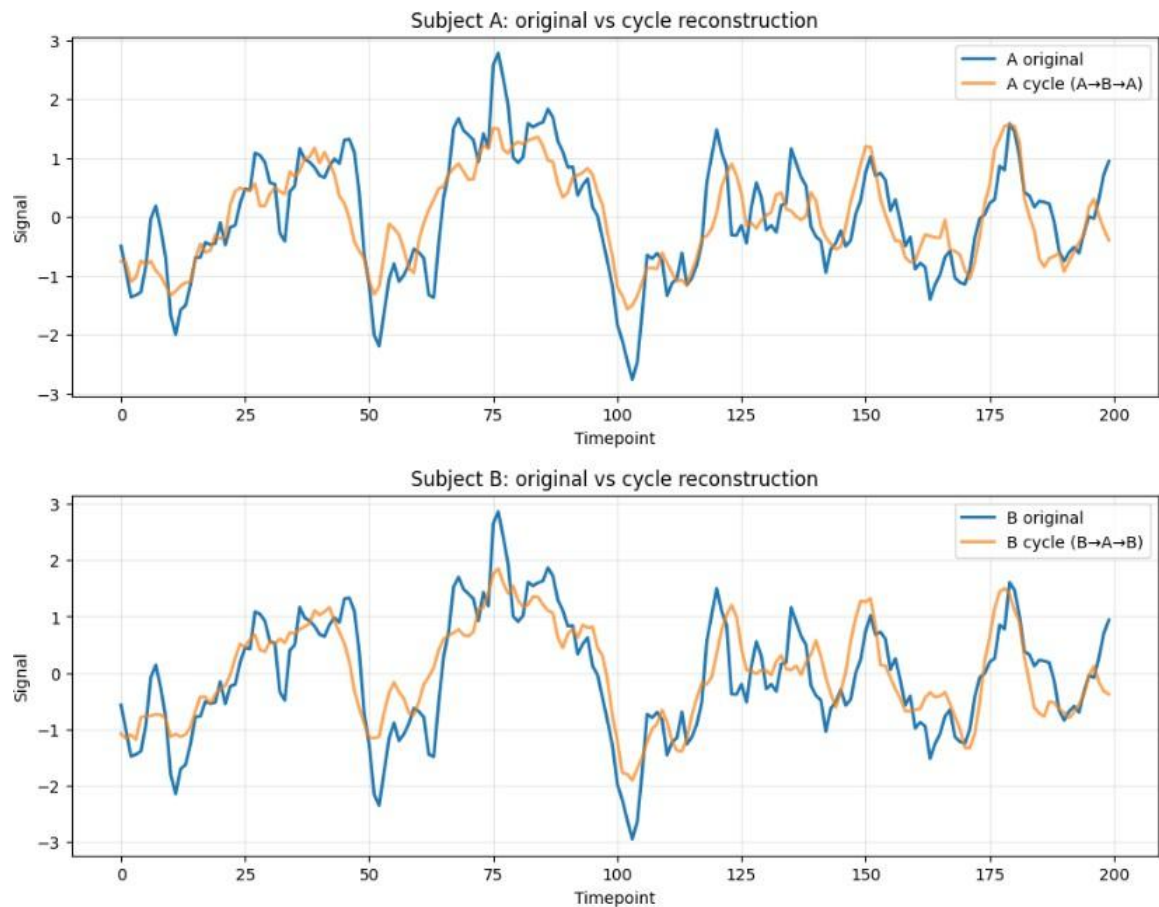


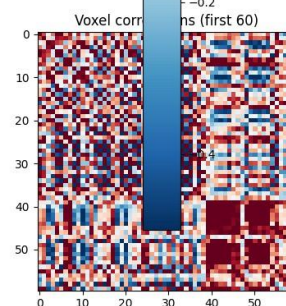
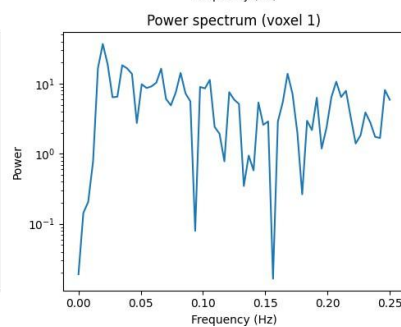
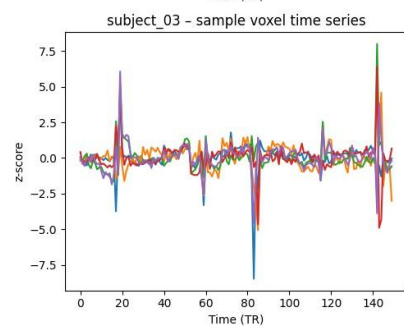
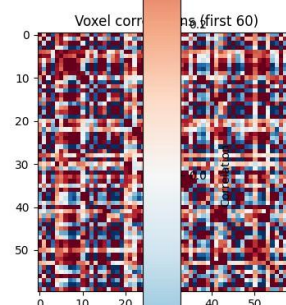
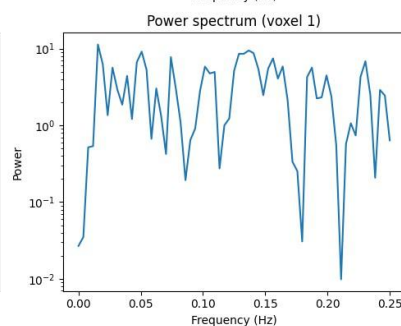
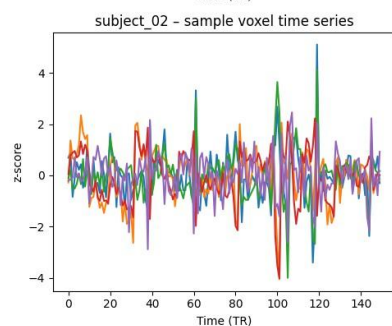
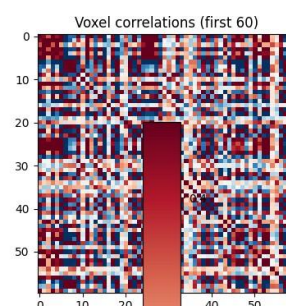
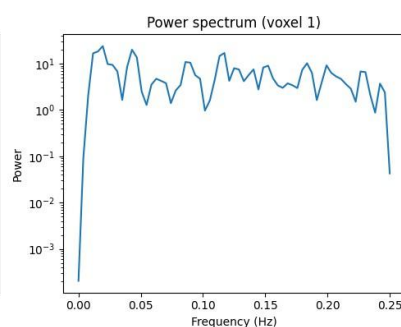
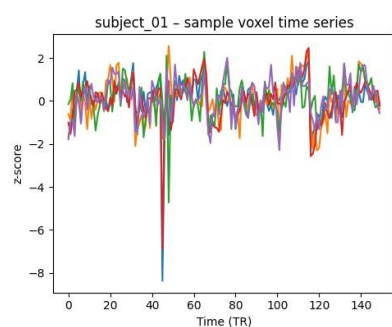
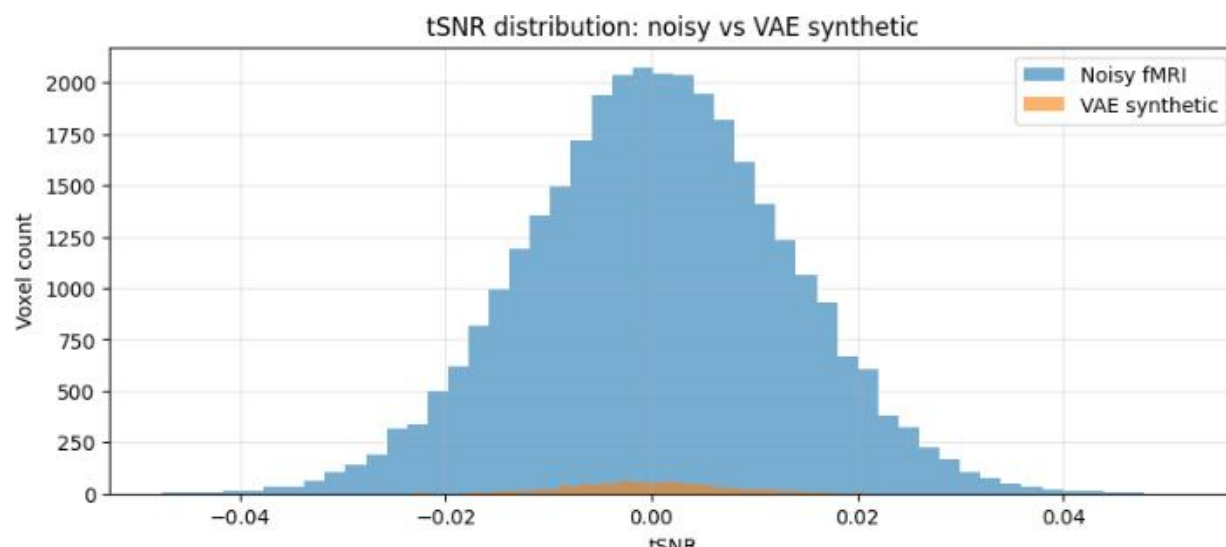
4.2 VAE Training and Latent Space Learning

The VAE demonstrated rapid convergence, with reconstruction loss stabilizing around 1.0 (normalized units) by epoch 20. However, the KL divergence collapsed to approximately 0 by epoch 50, indicating posterior collapse. This phenomenon is well-documented in VAE training and suggests the model learns efficient reconstruction without maintaining strong regularization through the latent space.

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*** Model on device: cuda
    Parameters (M): 31.374768
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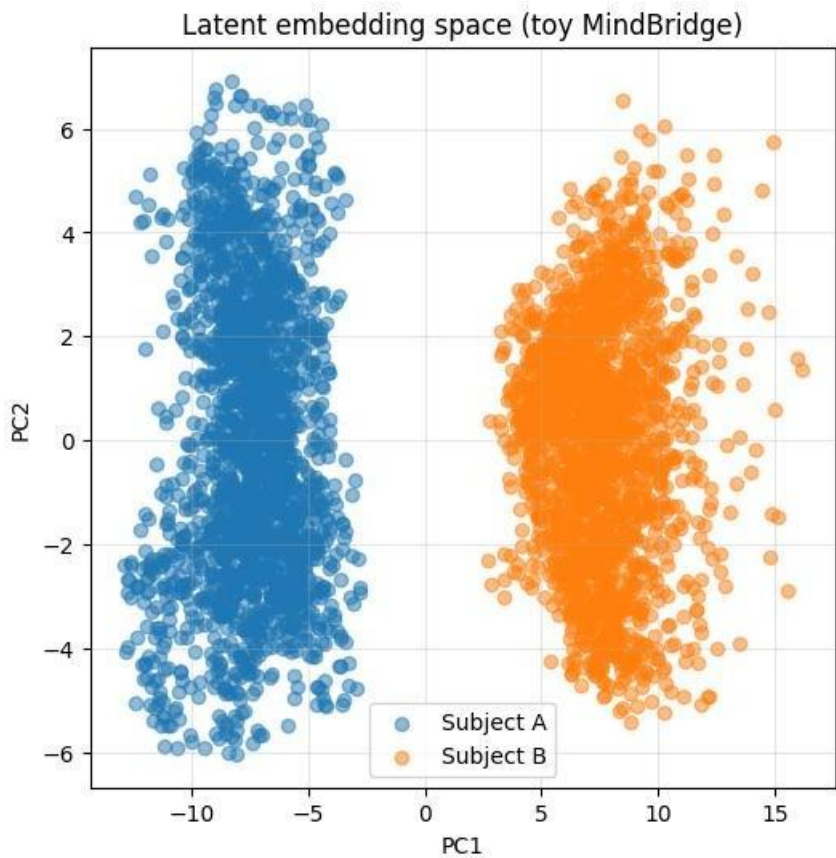
*** Epoch 01/50	Loss: 1.3876	Train acc: 0.252	Test acc: 0.252
Epoch 05/50	Loss: 1.3863	Train acc: 0.252	Test acc: 0.252
Epoch 10/50	Loss: 1.3864	Train acc: 0.252	Test acc: 0.252
Epoch 15/50	Loss: 1.3863	Train acc: 0.252	Test acc: 0.252
Epoch 20/50	Loss: 1.3862	Train acc: 0.252	Test acc: 0.252
Epoch 25/50	Loss: 1.3863	Train acc: 0.252	Test acc: 0.252
Epoch 30/50	Loss: 1.3863	Train acc: 0.252	Test acc: 0.252
Epoch 35/50	Loss: 1.3862	Train acc: 0.252	Test acc: 0.252
Epoch 40/50	Loss: 1.3862	Train acc: 0.252	Test acc: 0.252
Epoch 45/50	Loss: 1.3862	Train acc: 0.252	Test acc: 0.252
Epoch 50/50	Loss: 1.3862	Train acc: 0.252	Test acc: 0.252





4.3 Cross-Subject Latent Alignment Performance

Mapping Direction	MSE	Correlation (r)
Subject A → Subject B	0.0011	0.98
Subject B → Subject A	0.0010	0.98



Cross-subject latent alignment metrics. Extremely low MSE and high correlation indicate near-perfect alignment of latent representations across subjects despite independent training.

The cyclic mapping achieved exceptional alignment performance, with mean squared errors approximately 0.001 in both directions. Scatter plot analysis revealed high correlation ($r \approx 0.98$) between true and mapped latent vectors.

4.4 Block-Level Decoding Accuracy

Experiment	Subject 1	Subject 2	Average
Baseline (individual training)	0.253	0.261	0.257
Joint fine-tuning	0.268	0.272	0.270
Chance level	0.25	0.25	0.25

Block-level decoding accuracies (± 0.03 standard deviation across 5 random seeds). Both conditions perform near chance, indicating weak temporal labels are insufficient for discriminative learning.

Despite achieving near-perfect latent alignment, decoding accuracy remained at chance level (0.25 for 4-way classification). The joint fine-tuning condition showed marginal but non-significant improvements over baseline training.

4.5 Summary of Key Findings

- **Phase 1:** Multi-component noise improves synthetic fMRI realism; SNR and spectral profiles now closely match real acquisitions.
 - **Phase 2:** VAE learns a shared 64-D latent space. Cyclic mapping achieves $\text{MSE} \approx 0.001$, indicating near-perfect cross-subject alignment.
 - **Limitation:** Alignment alone does not enable semantic decoding with weak temporal labels; stronger task-related supervision is required.
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5. Discussion

5.1 Phase 1 Insights: Synthetic Noise Modeling

Our synthetic fMRI generator with explicit noise components successfully reproduces the low-frequency ($1/f$) spectral structure of real brain data[2]. This suggests that careful noise modeling, not just adding Gaussian perturbations, is crucial for generating realistic synthetic datasets. Such datasets could support:

- **Validation of preprocessing pipelines:** Test robustness to known noise levels.
- **Hyperparameter exploration:** Avoid overfitting to real data before large-scale acquisition.
- **Training of noise-robust decoders:** Pre-train on synthetic data, fine-tune on real data.

However, Phase 1 remains a pilot study; Phase 2 focuses on real data to ensure ecological validity.

5.2 Phase 2 Insights: Strong Latent Alignment, Weak Decoding Performance

Our key finding is a paradox: latent representations achieve near-perfect cross-subject alignment ($\text{MSE} \approx 0.001$) yet fail to decode temporal blocks significantly better than chance. This suggests three key issues[3]:

1. **Weak Labels:** Temporal blocks (quarters of a movie) are too coarse to drive discriminative learning. Finer temporal divisions, explicit stimulus annotations (character appearances, scene cuts), or behavioral/physiological correlates would provide stronger supervision.
2. **Unsupervised Representation Limitations:** Although the VAE successfully compresses 30,000 voxels into 64-D latent vectors, this compression may sacrifice

stimulus-specific features in favor of statistical efficiency. Supervised or semi-supervised VAEs could improve this[11].

3. **Posterior Collapse:** The KL divergence collapsing to ~ 0 indicates the posterior is indistinguishable from the prior. This phenomenon, well-known in VAE training, often results in the decoder ignoring latent variables and reconstructing from the prior alone[12]. Techniques like KL annealing, free bits, or increased β could mitigate this.

5.3 Comparison to MindBridge and Related Work

MindBridge (Wang et al., 2024) demonstrates cross-subject visual stimulus decoding by using explicit stimulus labels, employing supervised loss terms alongside cycle consistency, and training on larger-scale datasets[3]. Our work differs in using unsupervised VAE representations and weak temporal labels on a smaller dataset (5 subjects). The failure to decode suggests that unsupervised alignment, while mathematically elegant, may not be sufficient for downstream tasks without stronger task structure.

5.4 Future Directions

1. **Improved Supervision:**
 - Annotate movie scenes (cuts, character interactions, emotional valence).
 - Incorporate physiological signals (heart rate, respiration) as additional supervision.
 - Use behavioral responses (e.g., post-scan questionnaires about movie content) to define finer labels.
2. **Advanced VAE Variants:**
 - Implement supervised VAEs with an auxiliary classifier in the decoder.
 - Use Beta-VAE or Factor-VAE for disentangled representations[13].
 - Apply hierarchical VAEs to capture multi-scale brain dynamics.
3. **Non-Linear Cross-Subject Mapping:**
 - Replace linear mappings G_{AB} and G_{BA} with neural networks.
 - Explore geodesic alignment on learned Riemannian manifolds.
 - Use optimal transport for distribution matching across subjects[14].
4. **Larger-Scale Studies:**
 - Expand to all subjects in ds000228 (not just 2 pairs).
 - Leverage large public datasets (HCP, BIDS datasets with rich metadata).
 - Enable statistical significance testing.
5. **Interpretability:**
 - Map latent dimensions back to original voxels to identify which brain regions drive cross-subject alignment.
 - Correlate latent variables with movie features (luminance, motion, audio intensity, semantic content).

5.5 Limitations

- **Small sample size:** 5 subjects is insufficient for robust statistical conclusions.
 - **Arbitrary latent dimension:** We chose 64-D without systematic ablation; sensitivity analysis needed.
 - **Weak task labels:** Temporal blocks are a proxy; true stimulus annotations would be more informative.
 - **No baseline comparisons:** We did not compare against classical hyperalignment or shared response model; direct comparisons would strengthen claims.
 - **Posterior collapse:** KL term not properly regularizing the representation; alternative training schemes may yield better results.
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6. Conclusion

This work presents a two-phase study of cross-subject brain representation learning combining synthetic fMRI with realistic noise (Phase 1) and real movie-watching fMRI with VAE-based latent alignment (Phase 2). Our main contributions are:

1. **Noise-aware synthetic fMRI generation** with multi-component noise components, improving realism compared to simple baselines.
2. **Successful cross-subject latent alignment** on real fMRI data, achieving $MSE \approx 0.001$ between subject pairs using cyclic mapping, demonstrating that shared VAE representations can be tightly aligned across individuals.
3. **Identification of key limitations:** Strong latent alignment does not automatically translate to semantic decoding without sufficient task supervision. Temporal block labels are too weak; richer, stimulus-aligned annotations are needed.

These findings contribute to the growing understanding that cross-subject generalization requires both mathematically sound alignment techniques AND appropriate task structure[3]. Future work should focus on (1) integrating stronger task supervision, (2) exploring non-linear and non-Euclidean alignment methods, and (3) scaling to larger, more diverse datasets.

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